

A Branch-Safe Spectral and Scattering Atlas for a Side-Coupled Fano–Anderson Impurity

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Abstract

We solve the infinite nearest-neighbour chain with one discrete state side-coupled at the origin. A physical-sheet Schur function gives exactly two simple bound states for every nonzero coupling, one on each side of the free band. Squaring the secular equation can create extra real roots; an explicit sign filter prevents their promotion to eigenvalues.

1 Frozen Hamiltonian and physical sheet

Let

$$\mathcal{H} = \ell^2(\mathbb{Z}) \oplus \mathbb{C}|d\rangle, \quad J > 0, \quad \varepsilon \in \mathbb{R}, \quad g \in \mathbb{R},$$

and define

$$(Hu)_n = J(u_{n+1} + u_{n-1}) + gu_d \delta_{n0}, \quad (1)$$

$$(Hu)_d = \varepsilon u_d + gu_0. \quad (2)$$

This is a bounded self-adjoint operator on all of $\mathcal{H}$. Write H_0 for the free chain. Fourier transform makes H_0 multiplication by $2J \cos k$, so

$$\sigma(H_0) = \sigma_{\text{ac}}(H_0) = [-2J, 2J]$$

with multiplicity two in the open band.

Fix the analytic branch

$$S(z) = \sqrt{z^2 - 4J^2}, \quad S(z)/z \rightarrow 1 \quad (z \rightarrow \infty).$$

Thus $S(E) > 0$ for $E > 2J$ and $S(E) < 0$ for $E < -2J$. A contour integral, with the normalization checked at infinity, gives

$$m(z) = \langle 0, (z - H_0)^{-1} 0 \rangle = \frac{1}{S(z)}. \quad (3)$$

Block inversion at $|d\rangle$ now yields

$$G_{dd}(z) = \langle d, (z - H)^{-1} d \rangle = \frac{1}{D(z)}, \quad D(z) = z - \varepsilon - \frac{g^2}{S(z)}. \quad (4)$$

Theorem 1 (Two physical-sheet bound states). *If $g \neq 0$, H has exactly two exterior eigenvalues. They are simple and are the unique solutions*

$$E_+ - \varepsilon - \frac{g^2}{\sqrt{E_+^2 - 4J^2}} = 0, \quad E_+ > \max(2J, \varepsilon), \quad (5)$$

$$E_- - \varepsilon + \frac{g^2}{\sqrt{E_-^2 - 4J^2}} = 0, \quad E_- < \min(-2J, \varepsilon). \quad (6)$$

Squaring produces

$$(E - \varepsilon)^2(E^2 - 4J^2) - g^4 = 0, \quad (7)$$

but a real root of (7) is an eigenvalue only when it also satisfies the corresponding inequality in (5) or (6).

Proof. For $E > 2J$ put

$$D_+(E) = E - \varepsilon - \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at $2J+$ and to $+\infty$ at infinity, while

$$D'_+(E) = 1 + \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

Hence it has exactly one simple zero. The equation itself forces $E_+ > \varepsilon$. Below the band use

$$D_-(E) = E - \varepsilon + \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at minus infinity and to $+\infty$ at $-2J-$, and

$$D'_-(E) = 1 - \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

The unique zero is simple and satisfies $E_- < \varepsilon$.

At an exterior zero, equations (1)–(2) give the square-summable vector $u_{\mathbb{Z}} = gu_d(E - H_0)^{-1}|0\rangle$. Conversely, an exterior eigenvector cannot have $u_d = 0$, because H_0 has no exterior eigenvalue; its impurity equation therefore gives $D(E) = 0$. This establishes the pole/eigenvalue equivalence. Squaring loses the sign of $S(E)$ and the sign of $E - \varepsilon$, which proves the stated filter. $\square$

The phrase *branch-safe Schur owner* marks the baseline manuscript: the quartic is an algebraic candidate ledger, never an unfiltered spectrum.